



EXPLORERS LEVEL · GRADES K-1

Curriculum Guide

4 Semesters + Capstone · Complete Teaching Document

TEACHER EDITION

Scope & sequence · lesson plans · theory · projects · quizzes · alignment ·
inclusive design

1. Scope & Sequence

Explorers is the launch level: children in Kindergarten and Grade 1 meet coding for the very first time through **CODEship Blocks** — a visual block language built for emerging readers — with unplugged games and "peek at real code" moments along the way. Across four semesters and a capstone they learn that **code is instructions we give in order — and we can use it to help, create, and solve**. Every semester weaves the five CODEship strands — coding, math, English, cybersecurity, and AI literacy — around a real-world project. Reading is *emerging* at this age, so every instruction is low-text, audio/visual-first, and read aloud. Text-based coding begins at Builders; robotics is excluded at this level, and AI & cyber appear as age-appropriate seeds only.

Semester	Project	Coding focus	Big idea
1	My Helpful Robot	Sequencing; Move blocks; Start on Tap / Green Flag	Code is instructions in order — a robot can do a helpful job
2	Kindness Cards	Triggers (Start on Tap); Say; Sound; Go to Page	Screens can carry kindness; be kind online like in person
3	Recycling Sorter	Start on Bump; feedback; counting correct sorts	Computers can check and decide; code helps good choices
4	My Neighbourhood Map	Start on Tap on places; Say facts; Move navigation	Code can show my real community; keep home details private
Capstone	Explorer's Show-and-Tell App	Integrates sequence, triggers, feedback, click-a-place	I can use everything I learned to make something that teaches

How each semester runs

Each semester is **4 weekly classes** that build to **1 project**, and contains a **theory unit, 4 worksheets** (one per class), a **project planner**, and **2 quizzes of 10 questions** (Quiz 1 formative mid-block; Quiz 2 summative pre-project) with a warm answer key. The **capstone** is also 4 classes and is the **graduation gate** to Builders. Per level: 20 sessions · 5 projects · 8 quizzes · 4 theory units.

Block set used at this level (CODEship Blocks): Triggering — *Start on Green Flag, Start on Tap, Start on Bump, Start on Message*; Motion — *Move Right/Left/Up/Down, Turn, Hop, Go Home*; Looks — *Say, Grow, Shrink, Hide, Show*; Sound — *Pop, Record*; Control — *Wait, Repeat, Set Speed, Stop*; End — *End, Repeat Forever, Go to Page*. (Text-based coding begins at Builders.)

Session flow (same every class — predictability supports young & neurodiverse learners): Circle warm-up → Teach + theory (short) → Block-coding activity → Play & practice → Reflection/share.

2. Inclusive Design Playbook

Level-wide — applies to every lesson.

UDL baseline (all learners)

Representation: every instruction shown + spoken + pictured; read everything aloud; icon labels on blocks. **Engagement:** real-world hooks, choice of sprite/theme, movement, short segments. **Expression:** show learning by building, telling, drawing, or pointing — not only writing.

Accommodation profiles (K-1 specifics)

ADHD: 4 short segments per class; movement breaks; fidget-ok; one instruction at a time; instant feedback; visual checklist.

ASD: identical class routine; visual schedule card; explicit "what done looks like"; sensory-aware (sound/lighting); partnering optional.

Dyslexia / dysgraphia: audio + picture first; near-zero required reading/writing; speech or pointing to answer; never penalize spelling; colour + symbol coding.

Anxiety / low-confidence: pre-started templates; "mistakes help us learn" language; share to a partner, not the class; guaranteed early small win.

Gifted / 2e: each class has an "Explorer Extra" extension (extra step, second sprite, add a sound/loop); autonomy; no busywork.

3. Semester 1 — "My Helpful Robot"

Real-world hook: a robot that helps with a daily chore. **Coding focus:** sequencing + Move blocks + Start on Tap / Start on Green Flag (trigger blocks). **Big question:** *Can we give a computer instructions, in order, to do a helpful job?*

Theory unit

"Code is instructions in order." What an instruction is; why order matters (unplugged: "make a jam sandwich" gone wrong); computers only do exactly what we say. Gentle AI seed: "smart helpers like voice assistants follow instructions too."

Lesson plans

Class 1 — Instructions in order (unplugged + first blocks). Obj: put steps in sequence; place first blocks. Align: Math C3 sequential events; Language oral. Warm-up: act out "get ready for school" steps out of order — fix them. Teach + theory: order matters. Block activity: drag 3 Move blocks (Move Right/Up, Hop) so the robot travels to the toy. Practice: reorder blocks to change the path (worksheet). Reflect: what happened when order changed? Profiles: steps as picture cards; ADHD — one block at a time. Extra: add a 4th step. Assess: exit — point to "what comes first." Home: tell a grown-up 3 steps of a chore, in order.

Class 2 — Trigger blocks: Start on Tap / Start on Green Flag. Obj: start code with a trigger block. Warm-up: "when I clap, you freeze" (event game). Teach: "Start on Tap" (tap the robot) or "Start on Green Flag" starts the code. Block activity: Start on Tap → the robot uses Say to say "I'll help!" and moves. Practice: change what it says. Reflect: what starts the code? Profiles: anxiety — starter provided. Extra: add a sound. Assess: which block starts it? Home: draw your helpful robot.

Class 3 — Make it do the chore (sequence a task). Obj: sequence 4–5 blocks into a chore animation. Align: Math C3 sequence; Data (a "job" = ordered steps). Block activity: robot travels to the toys (Move/Hop) → Say "I'll help!" → moves to the bin (the visual block set has no pick-up block; show helping with Say/Grow). Practice: add one more helpful action. Reflect: which real chore did you code? Profiles: dysgraphia — no writing. Extra: second chore. Assess: robot completes the sequence.

Class 4 — Test, fix, share. Obj: find & fix one bug; present. Teach: "debug" = find the mistake. Activity: fix a broken robot (teacher-planted bug); finish own. Reflect + share: show a partner. Profiles: ASD — share to one partner; confidence — celebrate any fix. Assess: working robot + names one fix. Home: show family at next visit.

Sample worksheets

Class 1 — "Put It In Order": picture cards of a chore (1) pick up, (2) walk to bin, (3) drop in — child numbers or cuts-and-pastes into order; draws one more step. Others: C2 "Which block starts it?" match; C3 "My chore steps" draw-and-order; C4 "Spot the bug" circle-the-mistake.

Project outline

My Helpful Robot. Brief: code a robot that does one real chore. Success: uses a trigger block (Start on Tap / Green Flag) + at least 3 ordered Move/Looks blocks; robot completes a helpful task; child can tell what it does. Steps: pick a chore → order the blocks → add a Say block → test. Extension: add a sound or a second chore.

Rubric (score 1-4 each)

Starts on tap / green flag

Steps in order

Does a real chore

Can explain it

Ontario alignment

Math C3.1-C3.2 Coding (Gr 1): sequential events

Language: oral communication

Strand A: SEL

4. Semester 2 — "Kindness Cards"

Real-world hook: an interactive card that spreads kindness. **Coding focus:** trigger blocks (Start on Tap) + Say + Sound (Pop/Record) + Go to Page (scenes). Cyber/English seed: kind words online; "be kind on screens." **Big question:** *Can code help us share good feelings?*

Theory unit

"Screens can carry kindness." Words matter online too; kind vs. unkind messages; a first digital-citizenship seed — "we're kind on screens, like in person"; ask a grown-up if something feels wrong.

Lesson plans

Class 1 — What makes a kind message? Obj: name kind words; build a card sprite that says one. Align: Language oral/vocabulary; Technology & Society (kindness online). Warm-up: sort word cards kind/unkind (SEL). Block activity: Start on Tap → card uses Say for a kind message. Extra: add a second message. Profiles: dyslexia — pick words from pictures.

Class 2 — Add feeling with sound & colour. Obj: page/background change (Go to Page) + Sound on tap. Align: Math (patterns of colour); Arts cross-link. Activity: tap → Go to Page (new colours) + Pop/Record sound. Extra: a colour pattern (loop seed).

Class 3 — Make it interactive (trigger blocks). Obj: multiple tappable characters, each with Start on Tap. Align: Math C3 (triggers/events). Activity: tap different characters → different kind messages (Say). Profiles: ADHD — one part at a time.

Class 4 — Who's it for? Personalize + share. Obj: choose a recipient; test; present. Align: Language (audience). Reflect: who will you send kindness to? Profiles: anxiety — template card provided.

Sample worksheets

Class 1 — "Kind Words": circle the kind words; draw a face for how a kind message makes someone feel. Others: C2 colour-pattern colouring; C3 "match the tap to the message"; C4 "who is my card for?" draw.

Project outline

Kindness Cards. Brief: build an interactive digital card that shares kindness. Success: uses triggers (Start on Tap) + at least 2 kind messages + a sound/colour change; child names who it's for.

Rubric (score 1-4 each)

Interactive

2+ kind messages

Adds feeling (sound/colour)

Has an audience

Ontario alignment

Math C3 Coding (Gr 1): events

Language: vocabulary & audience

Technology & Society: kindness online

5. Semester 3 — "Recycling Sorter"

Real-world hook: a game that sorts items into the right recycling bins. **Coding focus:** Start on Bump (when characters touch) + feedback. Math/Data seed: sorting & categories; environment theme. **Big question:** *Can code help us make good choices, like recycling right?*

Theory unit

"Computers can check and decide." A rule = "if this, then that"; sorting into groups (data categories); why recycling right matters (real-world).

Lesson plans

Class 1 — Sorting is grouping. Obj: sort real items into categories (unplugged) → name the rule. Align: Math Strand D (sort/data), C3 (conditions). Activity: drag pictures into paper / recycling / compost. Extra: a 4th category.

Class 2 — "Start on Bump" gives feedback. Obj: item touches correct bin → "Yes!"; wrong bin → "Try again." Align: Math C3 (cause & effect); Start on Bump. Activity: build the check for one item. Profiles: confidence — starter provided.

Class 3 — More items, keep score of "good sorts." Obj: add items; count correct (number sense). Align: Math B (counting). Activity: add 2 more items + a happy counter. Extra: show the count.

Class 4 — Playtest & the "why." Obj: test the game; connect to real recycling. Reflect: one thing you'll recycle right now. Profiles: ASD — clear rubric + routine.

Sample worksheets

Class 1 — "Sort It Right": draw lines from items to the correct bin; write or say the rule for each bin. Others: C2 "if... then..." match; C3 tally the correct sorts; C4 "I can recycle..." draw.

Project outline

Recycling Sorter. Brief: a sorting game that rewards correct recycling. Success: at least 3 items; "if touching correct bin → positive feedback," else try again; a count/celebration for correct sorts; child explains the real-world point.

Rubric (score 1-4 each)

Sorts 3+ items

Gives right/wrong feedback

Counts good sorts

Explains why recycling matters

Ontario alignment

Math C3: conditions

Math Strand D: sort/classify data

Math Strand B: counting

Science: environment

6. Semester 4 — "My Neighbourhood Map"

Real-world hook: a clickable map of places the child knows. **Coding focus:** Start on Tap on multiple characters + Say facts + Move for simple navigation. Math/Community seed: position and spatial sense; places in my community. **Big question:** *Can code help me share where I live and what's around me?*

Theory unit

"Maps and place." Near/far, left/right/up/down (spatial); a map shows real places; a first "private vs. share" cyber seed — "we don't share exactly where we live with strangers online."

Lesson plans

Class 1 — My places (plan the map). Obj: name 3–4 places (home area, park, school, store); position on a grid. Align: Math E (spatial), Social Studies (community). Activity: place sprites on a backdrop map. Profiles: dyslexia — picture icons for places.

Class 2 — Click a place, learn a fact. Obj: Start on Tap on a place → it uses Say to tell a fact. Align: Math C3 events; Language (speaking facts). Activity: add facts to 2 places. Extra: add a sound per place.

Class 3 — Get around (simple navigation). Obj: move a character between places with Move blocks (position/direction). Align: Math E / C3. Activity: Move blocks (Right/Left/Up/Down) to move the sprite park → school. Profiles: ADHD — one route at a time.

Class 4 — Stay safe + share. Obj: the private-vs-share rule; present the map. Reflect: which places are okay to show? Profiles: ASD — explicit examples of ok / not-ok.

Sample worksheets

Class 1 — "Places Near Me": draw 4 places and label near/far, left/right. Others: C2 "a fact about my place"; C3 "map a route" arrows; C4 "safe to share?" sort.

Project outline

My Neighbourhood Map. Brief: an interactive map of real places you know. Success: 3 or more clickable places, each says a fact; a character can move between two; child follows the "don't share exact home details" rule.

Rubric (score 1-4 each)

3+ places

Click → fact works

Can navigate

Applies the safety rule

Ontario alignment

Math C3: events

Math Strand E: spatial sense (position/direction)

Social Studies: community

Cyber: private vs. share

7. Capstone — "Explorer's Show-and-Tell App"

Real-world hook: teach someone else one thing you learned. Integrates all four semesters: sequence (S1) + triggers & Say/Sound (S2) + a simple check/feedback (S3) + Start on Tap & Move positioning (S4). **Big question:** *Can I use everything I've learned to make something that teaches?*

Format: 4 classes (same as a semester) — plan → build → test → present. Passing graduates to Builders.

Theory unit

"I'm a coder now." Recap the big ideas (order, triggers, rules, sharing kindly & safely); "what will you teach?"

Lesson plans

Class 1 — Plan. Choose one thing to teach (a chore, kindness, recycling, or a place); storyboard 3 screens. Profiles: all — choice honours interest; templates for anxiety.

Class 2 — Build core. Trigger start (Start on Tap / Green Flag) + sequence + a Say that teaches. Extra: a sound.

Class 3 — Add interaction + a check. Clickable choice → feedback (uses the Semester 3 idea). Profiles: ADHD — checklist of parts.

Class 4 — Test & present. Debug; present to the group/parents at the showcase. Profiles: ASD/anxiety — present to one adult if preferred.

Sample worksheets

"My Show-and-Tell Plan": 3-box storyboard (what each screen teaches) + which blocks I'll use.

Project outline

Explorer's Show-and-Tell App. Success: uses ≥3 of the 4 learned skills (sequence, event, feedback, click-a-place); teaches one clear thing; child presents it and explains their code. The rubric below is the graduation gate.

Rubric — graduation gate (score 1-4 each)

Uses 3+ learned skills

Teaches something clearly

Interactive

Presents & explains

Kindness/safety awareness

Ontario alignment

Math C3: integrate sequence/events/conditions

Strand A: SEL

Language: presenting & explaining

8. Level Assessment, Graduation & Alignment

Assessment model

Formative: exit tickets + Quiz 1 in each block (warm, "let's review" feedback). **Summative:** Quiz 2 + the project rubric in each block. Every block ships a ready-to-use Quiz Pack (Quiz 1 · Quiz 2 · teacher answer key) and a landscape Assessment Sheet (rubric recording + quiz scores & observation).

Graduation to Builders

Complete the 4 semester projects + pass the capstone rubric (4 of 5 criteria met). The student then receives their certificate and levels up to Builders.

Alignment & compliance

Explorers is built on the pan-Canadian five-area spine (Programming; Computing & Networks; Data; Technology & Society; Design) and tagged first to Ontario. Coding maps to Mathematics Strand C3 (Coding), Grade 1 — sequential events and reading/altering code — with counting (Strand B), sort/classify data (Strand D), and spatial sense (Strand E) woven through the projects. Oral language, kindness online, and private-vs-share digital citizenship map to the Language curriculum and Technology & Society expectations at an emergent level. The inclusive-design playbook (§2) and the five accommodation profiles overlay every lesson.

Language (locked): always describe materials as "aligned to / maps to / supports" provincial expectations — never "endorsed by" or "approved by" a ministry. Validate each alignment against the current official curriculum documents before public use. Detailed provincial crosswalks (Ontario, BC ADST, Alberta, Québec + French) are maintained separately and are not part of this edition.